fat, fiber, gas producing and spicy foods. Sometimes very hot or very cold foods may irritate the intestines. Foods that are only warm or cool may be better accepted.

3. **Dry and sore mouth:** Patients may suffer from dry and sore mouth and may have problems with chewing and swallowing. Soft, bland and pureed foods are better tolerated. Frequent liquid intake is recommended. Using a straw instead of a spoon makes swallowing easier.

4. **Nausea and vomiting:** It may occur in any type of treatment. Eating dry foods such as crackers, toast, rice or other grain products, drinking beverage separate from meals, and serving foods cool instead of steaming hot are recommended.

12.10 Nutritional care in cancer

A well balanced nutritious diet is essential in order to maintain a healthy body weight and also to cope with the side effects of treatment procedures.

1. **Energy, carbohydrate, protein and fat:** The nutrient and calorie needs of a cancer patient are greater than they were before the onset of the disease. Tissues must be rebuilt and the nutrients lost to the cancer must be replaced. Therefore additional protein intake is required to rebuild tissues and to maintain a healthy immune system. Energy demands are high in cancer. Carbohydrates and fat will be needed to provide this energy and spare protein for tissue building and immune function. A high-protein and high calorie diet may be recommended during the treatment phase.

2. **Vitamins and minerals:** They are essential for metabolism and tissue maintenance. Vitamin and mineral supplements are provided to enhance immune function.

3. **Fluids:** Adequate fluid intake is required to keep the body hydrated and to maintain electrolyte balance. Fluids are important to help the kidneys eliminate the metabolic wastes and the toxins from drugs.

4. **Neutropenic diet:** A neutropenic diet is given to individuals with weakened immune systems. This diet helps to protect them from bacteria and other harmful organisms found in some food and drinks. In this diet, all raw fruits, vegetables and nuts are avoided. Fruits and vegetables may be consumed in the well-cooked form.

5. If chewing is a problem, a soft diet may be helpful. In case of diarrhoea, a low-residue diet may be recommended. A low residue diet limits the intake of fibre rich foods such as greens, whole grains, nuts and raw fruits and vegetables.

6. Oral feedings are preferred. However in extreme cachexia, enteral (tube feeding) or total parenteral (intravenous) feedings may become necessary.

Case Study

Mrs. R is a 45 year old woman who was diagnosed with stage III breast cancer. She underwent mastectomy (surgical removal of the whole breast). This was followed by chemo and radiation therapies. As an associated effect of the treatment procedures, she has developed mouth ulcers, diarrhoea and anorexia. What kind of dietary modification would you recommend to ensure nutritional adequacy?

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